PROJECT CADASTRAL CAUAYAN

— — — How borders aren't just ends — — —

PRE-PROJECT LOG

JULY 2026

CONFLICT TYPES

Conflict Type I = Historical Shift (since 2012-2014)

Conflict Type II = Estimation

Conflict Type III = Digitization Error

CLASSIFICATION

RT = Remote Tracing; needs only digital intervention—only shown because of GV upgrade possibility

GV = Ground Verification; needs extra fieldwork

DISCREPANCY CODES

- The letter at the beginning of the discrepancy code represents where the error is located (W for west of the Cagayan River, E for east of the river, and C if the discrepancy qualifies for both W and E)
- The letter is followed by a colon symbol then a number in the range of 1-5 to represent which barangay cluster the error is located, with
  - — 1 being Poblacion
  - — 2 being Tanap
  - — 3 being West Tabacal
  - — 4 being East Tabacal
  - — 5 being Forest (Region)
- If it involves two clusters, you may choose any of the two clusters' number involved
- It is then followed by a dash then another number, to differentiate the discrepancies in the same area from each other
- If it is finished, it is tagged with a [C] at the very beginning

EXTRA LAYERS

- Disconnected Areas are areas where it is disconnected to the barangay center/motherland
- Evidence is a Google Earth Pro (GEP) layer where it shows evidence of some discrepancies
- Cautionary Evidence is a GEP layer where it shows false/unlikely evidence
- Boundaries is the outdated map made by the CCPDCO

NEEDED FILES (INVOLVED LGUs)

- Property Identification Maps (PIMs)—Assessor's Office
- Section Index Maps—Assessor's Office
- Tax Declarations (TDs)—Assessor's Office Specifically boundary TDs, only to verify PIM discrepancies
- Boundary Computation Sheets—DENR-LMB May need assistance from the CSU to retrieve the file from the regional office

NEEDED EQUIPMENT (UP DGE) UP is not required to buy equipment if there is none at supply, although highly recommended to proceed

- GNSS RTK Base & Rover
- Tripods & Range Poles
- Electronic Total Station (ETS) + Prism Pole
- Rugged Handheld GPS
- Rugged Tablet + QField
- High-Capacity Power Banks
- Long Range Walkie-Talkies
- 50m Steel Tape (if signals are obstructed)
- Camera
- Field Gear

NEEDED SOFTWARE (OSMph)

- QGIS + Latest (ESRI) Imagery

GV DISCREPANCIES LIST

E:5-11

- IMPORTANCE: Low

E:5-33, 34, & 36

- IMPORTANCE: Low

E:5-4

- IMPORTANCE: High

E:5-5

- IMPORTANCE: High

E:5-6

- IMPORTANCE: High

E:5-7

- IMPORTANCE: High

W:1-17

- IMPORTANCE: Moderate

WORKFLOW PLAN

ROLES & RESPONSIBILITIES

- The handful volunteering OSMph team (or any determined) will be referred internally as simply the "Digital Team". Because it will likely compose of about 3-10 members, it will be an equal system upon the team with simply a layer of verification by one member of the team if there happens to be a human error made from trying to fix the boundaries.

- The Digital Team is only responsible for fixing the digital map, such as to trace, topology check, finalize fieldwork data, etc.
- The UP team will be called the "Field Team" internally and is only responsible to verify found GV discrepancies, through ways such as muhon-hunting, social interviews, surveying, etc. as well as retrieving needed files from the given offices.
- Project lead user cccr210 serves as an equal member towards the cooperation, primarily integrated to the Digital Team.
- The Field Team must prioritize High-Importance GV discrepancies first, followed by Moderate-Importance and Low-Importance GVs.

COMMUNICATION PROTOCOL

- All communication between the two teams will be funneled into a unified group, where the platform will be decided upon a majority agreement.
- cccr210 will notify any agencies within the LGU through the General Services Office.
- UP will handle all interactions of any source of government.
- Any major decisions brought to the Digital Team will be accepted upon a majority agreement.
- At least one member in the Digital Team ideally needs to be online during fieldwork to minimize problems and confusion during conduction.

PROPOSALS LIST

- It is proposed that the Field Team will stay in Brgy. Villa Concepcion instead of the Poblacion barangays for efficient fuel use towards the GV discrepancies on the Forest Region (5) cluster.
- It is proposed that the two teams will hold a brief orientation before conducting a survey on an area to minimize confusion and data problems.
- It is proposed that the found muhons will be photographed with the team that found it to be advertised in social media platform Facebook through the cccr210 page, where it could be reposted by other pages such as the UP's.
- It is proposed that a second repository is made to publish non-sensitive data publicly to ensure transparency upon the project.

FIELDWORK PLAN

- The fieldwork will be conducted in accordance with the 5th point in the Roles & Responsibilities subsection.
- The Field Team must locate and document for GV discrepancies through methods such as

social interviews where it is necessary.

- All fieldwork data will be photographed and documented through the Rugged Tablet via QField for the Digital Team to finalize.
- If the 1st point in the Proposals List is approved, the Field Team will operate from Brgy. Villa Concepcion to conduct GV discrepancies' surveys in the Forest Region (5) cluster.

TAGGING SYSTEM

- A Remote Tracing (RT) discrepancy may be upgraded into a GV if the base map for tracing does not match the current satellite imagery, does not have any data whatsoever, is covered, or any other reason that validates the need to conduct fieldwork.
- The upgrade must be validated by one member in the Digital Team before it is brought up to the Field Team.
- If upgrade is successful, its old tag will still be noted on the discrepancy catalog for reference purposes.

DATA HANDOFF

- Once fieldwork upon a GV is completed, all collected data gathered by the Field Team must be sent through not only the unified group, but a repository platform (e.g., GitHub, Google Drive, etc.) as well.
- If the fourth point in the Proposals List is approved, the Digital Team will filter out the non-sensitive data in the repository to be published in the public repository.
- If a member finds the collected data unclear or insufficient, another member will be asked first to clarify things.
- If the other member also finds the data unclear or insufficient, it would be notified to the Field Team.

SIGN-OFF PROCESS

- Once all discrepancies have been resolved and finalized, the Digital Team must thoroughly inspect the map to ensure no errors have been left unresolved.
- Once inspection is complete, the map is submitted to the CCPDCO to be merged with their map. If they do not respond through email within 5 office days, it would be immediately be brought up physically through the UP.
- Once agreement with the CCPDCO is made, the Digital Team and their GIS Team may work with each other to compare and finalize to merge the two maps.
- Upon finishing, the map may be brought up by not only the City of Cauayan's Sangguniang Panlungsod (SP), but involved municipalities as well through an online meeting ideally having one representative per LGU.

- Once the map has been legalized by the involved LGUs, the data will be exported to OSM through the Data License signed by the CCPDCO in September 24, 2025 (PHT) using JOSM to mass-import the data.

RISK LOG

- Fieldwork is likely to be delayed by occasional weather disruptions, as the rainy season arrives and could make conditions on the field worse, flood areas that have discrepancies, etc.
- Rugged terrain in East Cauayan may need extra work, as some areas of fieldwork have steep slopes that can risk health of the surveyors on the field.

NOTE: There are files included with the Log that shows further data to backup the Log.

NOTE to the CCPDCO: If you have not done so, please do not present the upcoming map to the Sangguniang Panlungsod at the current moment. I am currently working on an attempt to contribute to your data.

Created by a Proud Cauayeño

Cccr210 — OSM Mapper

A Humanitarian Project

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